SHAP-Based Explanations for Chronic Kidney Disease Risk Prediction: Toward Trustworthy Clinical AI

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Abstract

Chronic kidney disease (CKD) affects over 850 million people worldwide but CKD clinical risk prediction tools often lack interpretability, limiting their adoption in high-stakes medical decisions. We address this gap by integrating SHapley Additive exPlanations (SHAP) with calibrated machine learning to deliver accurate and transparent CKD predictions. Using a real-world dataset of 750 patients from Al-Ramadi Teaching Hospital, Iraq, we trained Random Forest (RF) and XGBoost models that achieved AUC > 0.99 and ∼97% accuracy. The global SHAP analysis highlighted urea, creatinine, and glomerular filtration rate as dominant predictors, aligning with nephrology knowledge. The local SHAP narratives provided case-level explanations for CKD-positive, negative, and borderline cases. Subgroup analysis confirmed consistent performance and explanation quality across sex and age, supporting fairness. We also developed a clinician-facing dashboard that visualizes patient risk scores alongside SHAP-derived explanations to aid diagnosis and referral. The contribution of this work lies in integrating calibrated tree-based models with global and local SHAP explanations, subgroup fairness assessment, and a clinician-oriented dashboard—an end-to-end framework that has not been previously applied to CKD laboratory data.

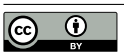
CCS Concepts

• **Applied computing**; • **Health informatics**; • **Computing methodologies**; • **Machine learning**; • **Supervised learning by classification**;

Keywords

Chronic Kidney Disease, Explainable AI, Machine Learning in Healthcare, SHAP

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1 Introduction

Artificial Intelligence (AI) is increasingly applied in healthcare to support disease detection and decision-making through machine learning (ML) models [1], [2]. Despite strong predictive performance, adoption in clinical practice remains limited because most high-performing models operate as black boxes [3], [4]. In medicine—where patient safety is at stake—accuracy alone is not sufficient; clinicians must also understand why a model reaches its decision.

CKD, a progressive and often silent illness affecting more than 850 million people worldwide [5], exemplifies this challenge. Standard laboratory indicators such as serum creatinine, blood urea nitrogen, and glomerular filtration rate (GFR) form the diagnostic foundation [6], and ML models trained on these features have shown high accuracy [7], [8]. However, such systems are often not explainable, which undermines clinician trust and raises concerns around ethical accountability [9]. Explainable AI (XAI) has emerged to address this problem [10], with SHAP offering a model-agnostic approach to quantifying feature contributions [12]. Nevertheless, few studies offer patient-level interpretability for CKD lab data, or explore fairness across demographic groups [11]. This paper addresses these gaps using SHAP. We investigate both global SHAP analyses (identifying which factors contribute most to CKD across the population) and local SHAP analyses (explaining individual patient predictions).

Traditional CKD diagnosis relies on rule-based criteria such as estimated glomerular filtration rate (eGFR) thresholds below 60 mL/min/1.73 m^2, often calculated using formulas like MDRD or CKD-EPI. While clinically accepted, these methods apply uniform cutoffs and lack individualized risk estimation. In contrast, machine learning models can provide probabilistic predictions tailored to patient-specific biomarker profiles.

Our contributions are:

1. A calibrated CKD prediction pipeline with state-of-the-art accuracy.
2. Global and local SHAP explanations aligned with nephrology knowledge.
3. Fairness analysis across demographic subgroups.
4. A prototype clinician dashboard for actionable interpretability.

The remainder of this paper reviews related work (Section 2), presents methodology (Section 3), reports results and fairness analyses (Section 4), discusses clinical implications (Section 5), and concludes (Section 6).

2 BACKGROUND AND RELATED WORK: A COMPARATIVE DISCUSSION

The adoption of ML in healthcare has heightened the demand for explainability, now recognized as both a regulatory as well as an ethical imperative [12]. This challenge is particularly important in nephrology, where predictions influence high-stakes decisions such as dialysis initiation and kidney transplantation [13], [23]. LIME provides local approximations [16], while gradient-based methods such as Integrated Gradients and SmoothGrad are primarily designed for imaging tasks, with limited applicability to structured clinical data [17], [18]. SHAP is model-agnostic, especially effective for tabular datasets [19] and consistently more stable than LIME [20]. This study leverages SHAP's strengths to support CKD prediction, where both global interpretability and individualized explanations are essential.

SHAP has been adopted across multiple medical specialties, including oncology [21], cardiology [22], critical care [23], neurology [24], and radiology [25]. While these applications demonstrate SHAP's versatility, they rarely assess subgroup-specific performance or provide visual tools optimized for clinical workflows. In nephrology, explainable CKD models are beginning to emerge: Katuwal et al. developed interpretable Random Forest models for CKD staging [26]; others used SHAP to identify factors contributing to metabolic syndrome [27] or proposed hybrid rule-based approaches [28]. Yet, these works frequently rely on curated datasets and omit subgroup evaluation. By applying SHAP to a real-world Iraqi CKD dataset, this study provides both global and local explanations while explicitly evaluating subgroup consistency—dimensions that remain underexplored in CKD-focused research.

Although sex and age disparities in CKD outcomes are well documented [14], explainability studies rarely evaluate whether interpretations remain stable across demographic groups. Recent surveys underscore subgroup robustness as a central criterion of trustworthy AI [6]. Clinical usability is another persistent gap: while some prototypes integrate SHAP into dashboards [29], few undergo systematic validation or incorporate human-centered design principles. In contrast, our dashboard unites global feature attribution with patient-level SHAP narratives, addressing usability, interpretability, and accuracy in tandem.

In summary, while SHAP has demonstrated value across diverse domains, CKD applications remain limited and fragmented. Prior studies often prioritize performance or fidelity but overlook fairness, real-world validation, and clinician-facing implementation. Our work addresses these gaps by (1) using a real-world dataset, (2) integrating both global and local SHAP explanations, (3) evaluating subgroup consistency, and (4) prototyping a clinician-facing dashboard for actionable interpretability. Table 1 presents a comparison of representative studies and highlights how our work advances the field.

3 METHODOLOGY

This section describes the dataset, preprocessing, machine learning models, evaluation metrics, explainability framework, and the clinician-facing dashboard. The pipeline was designed to combine strong predictive performance with interpretable outputs, supporting both clinical reliability and trust.

3.1 Dataset and Preprocessing

The dataset comprised 750 anonymized patient records from Al-Ramadi Teaching Hospital in Iraq, including demographics, laboratory results, and comorbidities (e.g., serum creatinine, blood urea, GFR, hypertension, diabetes). The binary outcome indicated CKD vs. non-CKD. Missing values were imputed using the median for each variable to reduce the influence of outliers. Categorical features such as gender and albumin presence were transformed using label encoding. All numerical features were standardized using z-score normalization to ensure comparability across scales. The dataset was partitioned into a 70% training set, a 15% validation set, and a 15% test set using stratified sampling to preserve the CKD/non-CKD distribution across splits. Table 2 summarizes the attributes, their types, and clinical relevance.

The anonymized dataset is publicly available on Zenodo at https://doi.org/10.5281/zenodo.16792307, with all identifiers removed in compliance with institutional approvals and GDPR/HIPAA guidelines. Missing values were imputed, continuous features normalized, and categorical variables encoded. Data were split into training (70%), validation (15%), and testing (15%) sets.

3.2 Machine Learning Models

We trained RF and XGBoost as the primary models, selected for their robustness and strong performance on structured clinical data. Decision Tree (DT), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) models were included as baseline comparators. All models were implemented using Python's scikit-learn and XGBoost libraries and executed in a Google Colab environment. Categorical variables (e.g., gender) were label-encoded, continuous variables were standardized using z-score normalization, and missing values were imputed using the median. RF was configured with n_estimators = 300 and random_state = 42, while XGBoost parameters—such as tree depth and learning rate—were adjusted through manual exploration. Hyperparameters for both primary models were selected through iterative manual tuning on the validation split to balance discrimination performance and limit overfitting, given the modest dataset size. Model probability outputs were calibrated to support clinically meaningful interpretation.

Table 1: Comparison of representative related works and our contribution

Study / Domain	Method	Strength	Limitation	This Work Advances
Katuwal et al. [26] (CKD)	RF	Interpretable staging	No fairness analysis	Adds subgroup fairness
[27] (Metabolic syndrome)	SHAP	Factor identification	No local explanations	Provides patient-level SHAP
[18], [29] (Dashboards)	SHAP visualization	Prototype dashboards	Not clinician-centered	Clinician-facing design
[6], [30], [31]. (Surveys)	Fairness in AI	Identify bias risks	No CKD focus	Fairness evaluation in CKD
Our Work (CKD)	RF, XGBoost + SHAP	High accuracy, global + local SHAP, fairness, dashboard	—	Jointly addresses performance, interpretability, fairness, and usability

Table 2: Attributes in CKD dataset. Continuous and categorical clinical features, including units and target classification labels.

Attribute	Name	Type	Attribute values (range/unit)	Description
Gender	Gender	Nominal	Male, Female	Patient gender
Age	Age	Numerical	1–90 years	Patient's age in years
WBC	White blood cell	Numerical	4–10 $\times 10^9$/L	White blood cell count
RBC	Red blood cell	Numerical	4–6.2 $\times 10^{12}$/L	Red blood cell count
HGB	Hemoglobin	Numerical	11–16 g/dL	Hemoglobin concentration in blood
HCT	Hematocrit test	Numerical	37%–54 %	Proportion of blood volume occupied by red cells
PLT	Platelet count	Numerical	150–450 $\times 10^9$/L	Platelet count in blood
Urea	Blood urea	Numerical	15–45 mg/dL	Urea nitrogen concentration in plasma
Crea	Serum creatinine	Numerical	0.6–1.2 mg/dL	Creatinine level (kidney function marker)
UA	Uric acid	Numerical	2.5–6.8 mg/dL	Uric acid concentration in urine
Glu	Random blood glucose	Numerical	70–115 mg/dL	Blood glucose concentration
HPT	Hypertension	Nominal	Yes, No	Presence of high blood pressure
DM	Diabetes mellitus	Nominal	Yes, No	Presence of diabetes mellitus
GFR	Glomerular filtration rate	Numerical	0–90+ mL/min/1.73 m^2	Kidney filtration efficiency
CKD	Chronic kidney disease	Target	CKD, No CKD	Target outcome

3.3 Evaluation Metrics

We trained RF and XGBoost as the primary models, selected for their robustness and strong performance on structured clinical data. Decision Tree (DT), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) models were included as baseline comparators. All models were implemented using Python's scikit-learn and XGBoost libraries and executed in a Google Colab environment. Categorical variables (e.g., gender) were label-encoded, continuous variables were standardized using z-score normalization, and missing values were imputed using the median. RF was configured with n_estimators = 300 and random_state = 42, while XGBoost parameters—such as tree depth and learning rate—were adjusted through manual exploration. Hyperparameters for both primary models were selected through iterative manual tuning on the validation split to balance discrimination performance and limit overfitting, given the modest dataset size. Model probability outputs were calibrated to support clinically meaningful interpretation.

Performance was assessed using Accuracy, Precision, Recall, Specificity, F1-score, and the Area Under the ROC Curve (AUC). Accuracy reflects the overall proportion of correct classifications. Precision measures the fraction of predicted CKD cases that are truly positive, whereas Recall quantifies the model's ability to detect CKD cases. Specificity captures the correct identification of non-CKD instances. The F1-score provides a harmonic balance between Precision and Recall, which is valuable when class distributions are uneven. AUC evaluates discrimination across all decision thresholds. These performance metrics were defined as:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives.

Calibration was examined using the Brier score, which measures the mean squared deviation between predicted probabilities and observed outcomes, and the Expected Calibration Error (ECE), which compares predicted probability levels with empirically observed event frequencies. Together, these metrics characterize discrimination performance, sensitivity to CKD detection, and the reliability of probability estimates.

3.4 Explainability with SHAP

To address the interpretability gap in machine learning models, we employed SHAP. It is based on cooperative game theory and decomposes the model output into additive contributions of individual features. This allows the prediction for a patient to be expressed as:

$$f(x) = E[f(x)] + \Sigma \varphi_i$$

where E[f(x)] is the expected model output across the dataset and φ_i is the SHAP value of feature i for input x.

This formulation provides both global and local interpretability. Global explanations, generated through SHAP summary plots, highlight the most influential features across the entire population. Local explanations, visualized with force plots and waterfall plots, reveal how individual patient features influence specific predictions. For example, high serum creatinine and low GFR typically appear as strong positive contributors to CKD predictions, consistent with clinical knowledge.

Subgroup analyses were also performed, comparing SHAP value distributions across age and gender. This ensured that explanations remained consistent across patient demographics, an important step toward fairness and trustworthy clinical deployment. SHAP was applied only after ensuring strong model calibration and discrimination, as indicated by high AUC values (>0.99) and low Brier scores (<0.03), to prevent misleading attributions.

3.5 Clinician-Facing Dashboard and Workflow Summary

The overall methodology can be summarized as a sequential process:

Data Collection → *Preprocessing*
→ *Model Training and Calibration* → *SHAP Explanations*
→ *Subgroup Fairness Analysis* → *Dashboard Integration*

This framework integrates predictive accuracy, interpretability, and fairness, offering a pathway toward trustworthy AI in healthcare. We developed a prototype dashboard to visualize model predictions alongside SHAP-based explanations. This interface is intended to support clinician interpretation and workflow integration. While usability testing is outside the scope of this study, the dashboard offers a foundation for future human-in-the-loop design and evaluation. The dashboard displays SHAP feature contributions as color-coded horizontal bars to separate risk-increasing from risk-reducing biomarkers, thereby reducing cognitive load for clinicians. The layout follows common decision-support conventions by presenting the predicted risk score alongside the most influential features contributing to the prediction. The methodological pipeline integrates predictive modeling, SHAP explanations, subgroup analysis, and dashboard visualization (Figure 1).

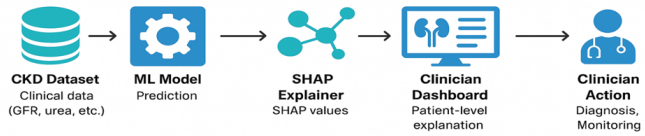


Figure 1: CKD Prediction and Explanation Pipeline. Clinical data from patients with CKD are processed through a ML model to generate predictions. SHAP values are then computed to explain model outputs at the patient level, which are visualized on a clinician dashboard to support diagnosis, referral, and monitoring decisions.

4 Results and discussion

While our models are standard classifiers, the integrated focus on explainability (via SHAP), fairness analysis, and clinical usability through a decision-support dashboard represents a novel contribution in the context of CKD risk prediction. All performance metrics reported are based on a held-out 20% test set that was not used during training or model selection, ensuring an unbiased evaluation.

4.1 Model Performance

We compared five models—RF, XGBoost, DT, SVM, and MLP—using the same preprocessing pipeline. Performance was evaluated on the held-out test set with Accuracy, Precision, Recall, F1-score, AUC, and Brier Score. The latter is particularly important in healthcare, as it reflects the reliability of probability estimates.

Table 3 summarizes the results. RF and XGBoost achieved the best performance, with Accuracy ≈ 96–97%, AUC > 0.99, and low Brier scores (0.027–0.029). SVM was competitive in accuracy and discrimination but showed weaker calibration (Brier ≈ 0.035). MLP reached 93% Accuracy but poorer calibration (0.059), while DT lagged significantly with 87% Accuracy and poor robustness (Brier 0.133). These findings establish RF and XGBoost as the most reliable models, balancing accuracy, discrimination, and calibration. Subsequent interpretability and subgroup analyses therefore focus on these two ensembles.

In addition to the summary metrics in Table 3, we examined how each model behaves on the test set using the fused confusion matrices shown in Figure 2. These plots make the differences between models much easier to understand. RF, XGBoost, and SVM all display nearly identical patterns: each model produces only a few errors, typically 2–3 false positives and 2–3 false negatives. This matches the high Recall values in Table 2 and confirms that these models rarely miss CKD-positive cases—a key requirement in a screening scenario.

The other two models behave differently. As per Figure 2, MLP makes more mistakes overall, and DT performs the weakest, with noticeably higher numbers of false positives and false negatives. These visual patterns help explain why these two models lag behind in accuracy and calibration. The DT's confusion matrix in particular highlights the model's difficulty in differentiating CKD from non-CKD cases, which is also reflected in its low AUC (0.838) and high Brier score (0.133). Complementing discrimination, the probability calibration was assessed through fused calibration curves (see Figure 3).

Table 3: Comparative performance of five machine learning models on the CKD dataset. RF and XGBoost achieved the best balance of accuracy, discrimination (AUC), and calibration (lowest Brier scores), followed closely by SVM.

Model	Accuracy	Precision	Recall	F1	AUC	Brier Score
Random Forest	0.967	0.982	0.973	0.977	0.995	0.027
XGBoost	0.960	0.973	0.973	0.973	0.991	0.029
SVM	0.960	0.973	0.973	0.973	0.990	0.035
MLP	0.933	0.955	0.955	0.955	0.969	0.059
Decision Tree	0.867	0.917	0.900	0.908	0.838	0.133

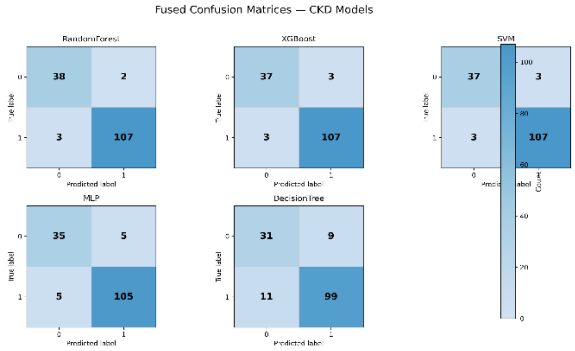


Figure 2: Fused confusion matrices for all five CKD models. Ensemble methods produce the most stable error profiles with minimal false negatives and false positives.

RF and XGBoost show curves that closely track the 45-degree reference line, consistent with their low Brier scores (0.027–0.029). This indicates that the predicted probabilities meaningfully correspond to true CKD risk—an essential requirement for deployment in decision-support settings. SVM also demonstrates acceptable calibration but with slight deviations at moderate probability levels, matching its higher Brier score (0.035). MLP presents under confidence in low- and mid-range probability bins, while DT exhibits substantial miscalibration throughout the probability range. These results reinforce the earlier conclusion that RF and XGBoost offer the best balance of discrimination and reliability. Importantly, well-calibrated models enable clinicians to interpret predicted risk scores in a probabilistically meaningful way, improving communication, appropriate prioritization of patients, and threshold-based referral decisions. Together, the fused confusion matrices and calibration curves provide a comprehensive view of classification reliability across the five models, supporting the selection of RF and XGBoost for the subsequent SHAP-based interpretability analyses in the next sub-sections.

4.2 Global SHAP Interpretability

To understand how RF and XGBoost arrive at their predictions, we applied SHAP at the global level. This analysis is crucial in clinical contexts, as it validates whether the models' logic aligns with established nephrology knowledge. Across both models, urea, creatinine, and eGFR consistently emerged as the strongest predictors of CKD, with hemoglobin and hematocrit also contributing (Figure 4). These results confirm medical plausibility, since they correspond to standard renal biomarkers and comorbidities.

Initial RF runs showed spurious emphasis on demographic variables (age, gender). After retraining with regularization and interventional SHAP, the feature rankings converged with those of XGBoost, emphasizing renal biomarkers over demographics. While both models highlight the same top predictors, RF distributes importance more broadly across hematological features, whereas XGBoost concentrates weight on fewer, highly discriminative variables (notably urea). Clinically, RF's broader feature distribution may support nuanced interpretation and monitoring, while XGBoost's sharper prioritization is better suited to fast screening applications.

Figure 5 shows the fused bar plots of mean absolute SHAP values, allowing direct comparison of feature importance between RF and XGBoost. Both models consistently rank urea, GFR, and creatinine among the top predictors, though RF distributes importance more evenly across renal and hematological features, while XGBoost assigns overwhelming dominance to urea. RF thus captures a broader spectrum of CKD risk factors, whereas XGBoost sharpens its focus on fewer but highly discriminative biomarkers.

From a clinical perspective, urea's prominence across both models reaffirms its established role in assessing renal dysfunction. RF's broader distribution of importance may support nuanced clinical interpretation and monitoring while XGBoost's sharper prioritization could be advantageous in screening applications where computational efficiency and high discriminative power are prioritized. In summary, global SHAP results show that RF and XGBoost are not only accurate but also clinically interpretable, reinforcing their suitability for downstream patient-level analysis.

4.3 Local SHAP Interpretability

While global SHAP analysis identifies the most influential features across the entire population, clinical decision-making often requires patient-level interpretability. Local SHAP explanations provide this by quantifying how each feature value contributed to the prediction for a given patient. This allows clinicians to not only see the model's final decision but also understand why the model labeled a patient as CKD-positive or CKD-negative. Figure 6 shows local SHAP decision plots for randomly selected patients predicted by RF and XGBoost

4.3.1 RF Results (Figure 6 – Left). CKD-Positive Patients: RF explanations consistently highlighted biochemical markers of impaired renal function:

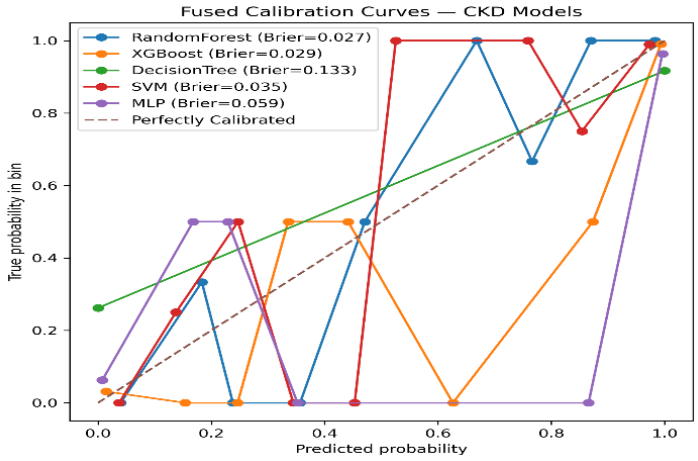


Figure 3: Fused calibration curves for all models. RF and XGBoost exhibit the closest alignment with perfect calibration, consistent with their low Brier scores

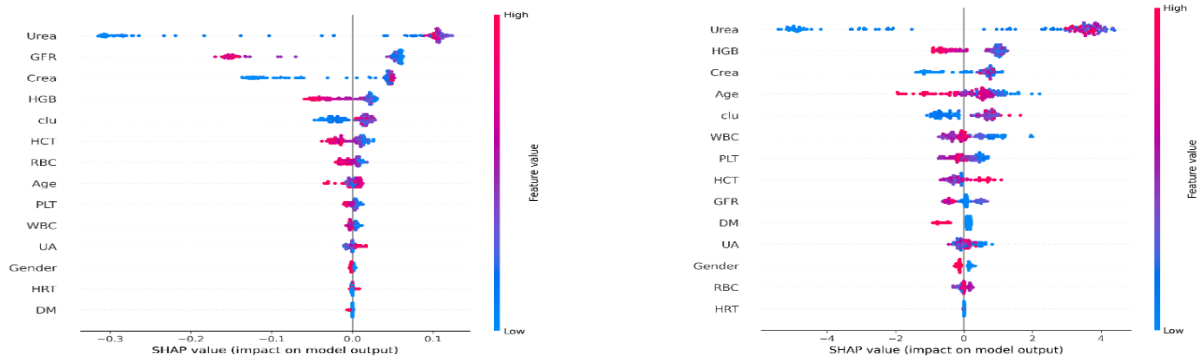


Figure 4: Global SHAP summary plots for (Left) Random Forest and (Right) XGBoost.

- Patient 88: High urea (56.7), elevated creatinine (2.13), and reduced eGFR (31) strongly pushed toward CKD. Minor protective pulls from hemoglobin and white blood cell count were insufficient to offset these dominant drivers.
- Patient 101: Severe renal impairment was evident, with extremely high urea (113), very low eGFR (11), and markedly elevated creatinine (5.1). Concomitant anemia (HGB 8.3, HCT 26) further reinforced the positive classification.
- Patient 1: CKD prediction arose from reduced eGFR (33) and moderately elevated creatinine (1.5), supported by low-normal hemoglobin.

CKD-Negative Patients: the opposite biochemical patterns dominated:

- Patient 36: Normal urea (21.7), preserved eGFR (75), and creatinine (1.04) pulled strongly away from CKD.
- Patient 37: eGFR (71) and creatinine (1.04) were protective, while normal hematological indices (HGB 14.9, RBC 5.0) reinforced the negative call.
- Patient 38: Very strong protection came from high eGFR (104) and low creatinine (0.8), leaving little doubt about non-CKD status.

The borderline Patient 3 showed a mixed profile: elevated urea (58) pushed toward CKD, but normal creatinine (0.8) and preserved eGFR (70) pulled strongly negative. The final probability remained near the threshold, reflecting a biologically plausible gray zone.

4.3.2 XGBoost Results (Figure 6 – Right). CKD-Positive Patients: XGB explanations were sharper, often dominated by urea:

- Patient 88: Urea (56.7) provided a large positive push, outweighing modest negatives from hemoglobin.
- Patient 101: Urea (113) alone nearly explained the CKD classification, with creatinine (5.1) and anemia adding reinforcement.
- Patient 1: Urea (113) and creatinine (5.1) pushed strongly positive, with age (67) and anemia contributing further.

CKD-Negative Patients: eGFR and creatinine played the strongest protective roles:

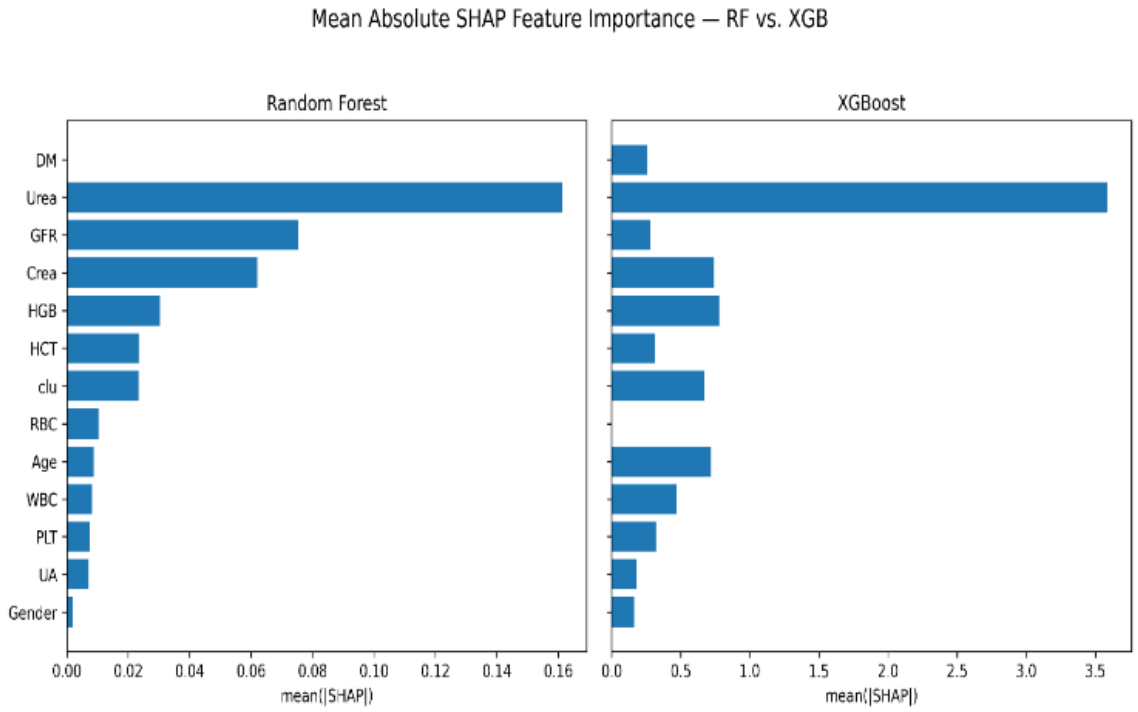


Figure 5: Mean absolute SHAP feature importance for Random Forest and XGBoost. The fused bar plots show the average magnitude of SHAP values for each feature, enabling direct comparison of their contribution to CKD prediction.

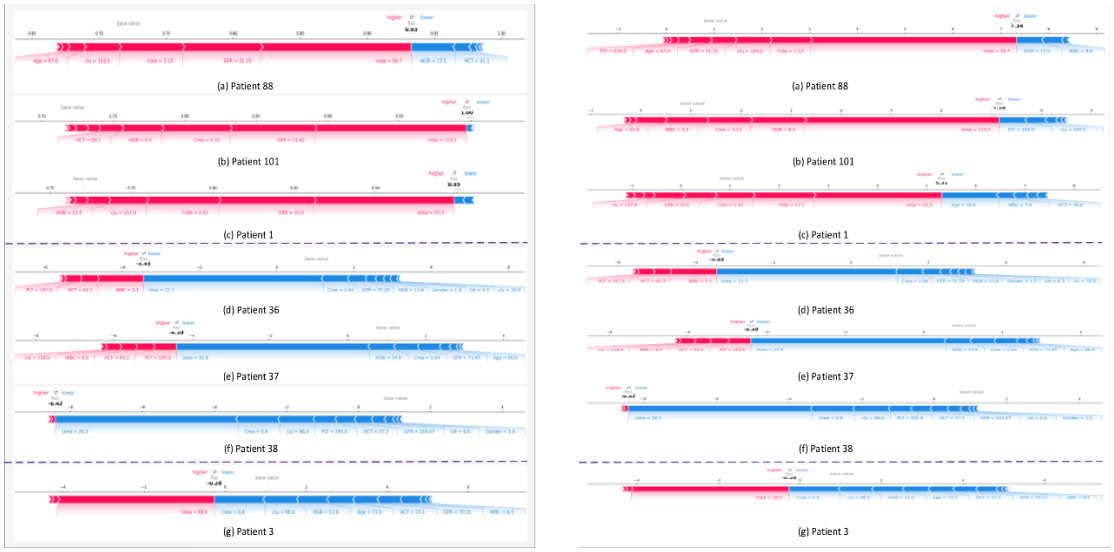


Figure 6: Local SHAP force plots for RF (top panels) and XGBoost (bottom panels). For each model, panels (a–c) show CKD-positive cases, panels (d–f) show CKD-negative cases, and panel (g) illustrates a borderline case. Red features increase predicted CKD risk, whereas blue features reduce it.

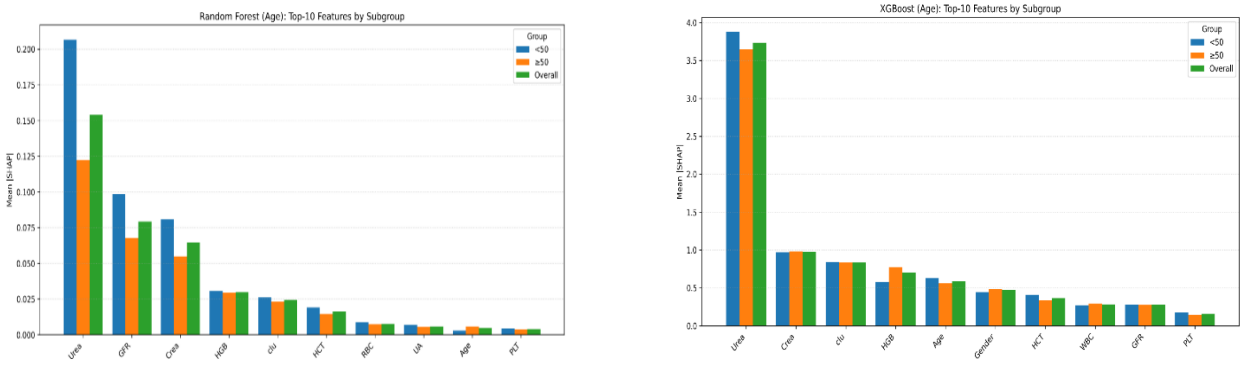


Figure 7: (Left) RF SHAP feature importance across age subgroups (<50 vs. ≥50). Urea, creatinine, and eGFR are consistently dominant, with minor shifts in relative influence. (Right) XGBoost SHAP feature importance across age subgroups (<50 vs. ≥50). Urea dominates across both groups, with creatinine and eGFR as consistent secondary drivers.

- Patient 36: eGFR (75) and creatinine (1.04) dominated, counteracting a mild urea contribution.
- Patient 37: Protective signals from eGFR (71), creatinine (1.04), and hemoglobin (14.9) clearly outweighed any red features.
- Patient 38: Normal renal function (eGFR 104, creatinine 0.8, urea 26.5) generated strong blue pulls, firmly classifying as non-CKD.

The borderline Patient 3 again illustrated conflict: urea (58) pushed toward CKD, while creatinine (0.8), eGFR (70), and hemoglobin (12.9) pulled away. The near-balance produced an uncertain probability, consistent with RF

4.4 Fairness Analysis

Fairness is essential in clinical AI, where predictive systems must not systematically favor or disadvantage specific groups. To evaluate fairness in CKD prediction, we examined RF and XGBoost across sex (male/female) and age (<50/≥50 years) subgroups. Two complementary aspects were assessed: (1) the stability of SHAP feature attributions across groups, and (2) comparative subgroup performance measured by AUC, sensitivity, specificity, and Brier score.

Across both models, urea, creatinine, and eGFR consistently emerged as the leading predictors in all subgroups, with only minor shifts observed in secondary features such as hemoglobin. Predictive performance remained uniformly high, with AUC values >0.95 and sensitivity close to 0.97 across all cohorts. The only systematic variation detected was a modest reduction in specificity and a small increase in Brier scores among older patients, which likely reflects underlying clinical heterogeneity rather than algorithmic bias. Importantly, sex showed no measurable influence on feature attributions or predictive outcomes, confirming that both models rely primarily on renal biomarkers rather than demographic shortcuts.

Taken together, these findings demonstrate that RF and XGBoost achieve subgroup fairness while maintaining complementary interpretability profiles: RF distributes predictive weight across a broader range of biomarkers, whereas XGBoost emphasizes a smaller set of dominant features. This complementarity strengthens confidence in their deployment, as both yield fair and clinically coherent predictions across demographic groups.

Figure 7 illustrates SHAP feature importance across age subgroups, where both models highlight urea, creatinine, and eGFR as dominant predictors with only minor differences in secondary contributions. Figure 8 presents SHAP feature importance across sex subgroups, showing nearly identical reliance on renal biomarkers for both male and female patients, with no evidence of sex-specific bias.

In assessing fairness, we found that key metrics such as AUC, sensitivity, and Brier score showed no significant variation between groups. SHAP value distributions and top features were consistent across subgroups, suggesting model behavior was stable and not biased toward any specific demographic.

Table 4 summarizes subgroup performance metrics. Consistent AUC, sensitivity, specificity, and Brier scores across sex and age groups indicate stable predictive behavior and no measurable demographic bias.

4.5 Discussions

The results from Sections 4.1–4.4 show that SHAP-based interpretability strengthens CKD risk prediction by combining high predictive accuracy with clinical plausibility and subgroup fairness. RF and XGBoost both achieved strong discrimination and reliable calibration, providing a robust foundation for model-assisted screening. Global SHAP analyses confirmed alignment with nephrology expertise, consistently ranking urea, creatinine, and eGFR as dominant predictors. At the individual level, local SHAP explanations generated coherent case narratives, including borderline patterns, allowing clinicians to scrutinize the reasoning behind each prediction. Subgroup analyses further demonstrated stable performance and explanations across sex and age, reducing concerns regarding demographic bias.

RF and XGBoost also offer complementary interpretability behaviors. RF distributes importance across a broader set of laboratory features, supporting detailed and context-rich clinical interpretation. In contrast, XGBoost concentrates its influence on a smaller

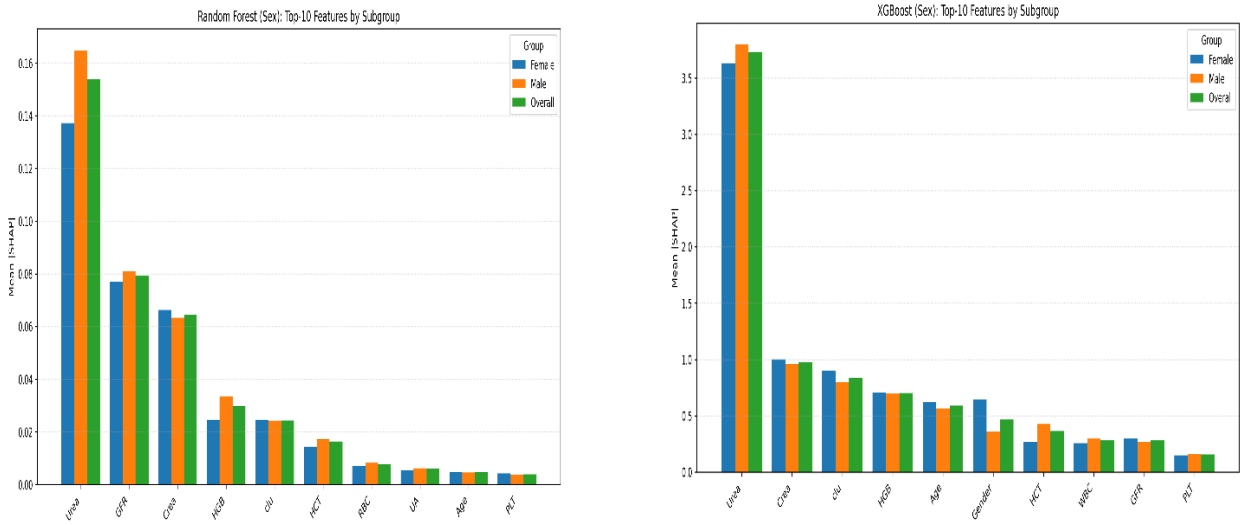


Figure 8: (Left) Random Forest SHAP feature importance across sex subgroups (male vs. female). Both subgroups rely on the same leading biomarkers, with no evidence of sex-specific bias.

Table 4: Subgroup performance summary

Subgroup	AUC	Sensitivity	Specificity	Brier Score
Male	>0.95	~0.97	High	Low
Female	>0.95	~0.97	High	Low
Age < 50	>0.95	~0.97	High	Low
Age ≥ 50	>0.95	~0.97	Slightly lower	Slightly higher

group of decisive biomarkers, enabling faster initial risk assessment and efficient identification of patients who may require further evaluation. This complementarity supports a practical two-stage workflow in which XGBoost provides rapid preliminary risk estimation, followed by RF for more in-depth clinical review when needed. The economic evaluation in Section 4.7 reinforces this perspective: both models demonstrated positive net clinical benefit across realistic decision thresholds and produced substantial projected savings through earlier CKD identification.

Despite these strengths, several limitations remain. SHAP values describe associations rather than causal effects, and explanation reliability depends on model fidelity and data quality. Fairness analyses were limited to sex and age; extending evaluation to additional dimensions such as comorbidity profiles or socioeconomic factors would further support equitable deployment. External validation across multiple clinical sites is needed to confirm generalizability beyond this dataset. Table 5 synthesizes performance, biomarker alignment, interpretability, and fairness, while Figure 9 illustrates how these elements contribute to a coherent framework for trustworthy CKD risk prediction.

4.6 Trust and Clinical Use

In clinical AI, predictive accuracy must be matched by interpretability to inspire trust. In this study, SHAP explanations bridged the gap by translating CKD predictions into clinically grounded narratives that support communication, referral, and monitoring. For physician–patient interaction, SHAP plots transform opaque probabilities into accessible explanations (e.g., elevated creatinine and reduced GFR driving high risk), improving shared decision-making. For referrals, SHAP highlights feature combinations that justify escalation beyond fixed thresholds. For longitudinal care, repeated SHAP evaluations reveal evolving risk drivers (e.g., rising urea), enabling early intervention. To demonstrate practical integration, we developed a prototype dashboard (Figure 10) that presents a patient's predicted CKD probability, referral suggestion, and top contributing features in a color-coded bar chart. This design reduces cognitive burden and aligns with familiar clinical tools while adding transparency. The dashboard layout is intended to support rapid interpretation by clinicians, although formal usability testing remains future work

Despite these benefits, SHAP has limitations: it is not causal, explanations depend on model and data quality, and human oversight remains essential. Nonetheless, by embedding interpretability into workflows, SHAP can support safer and more trustworthy AI in nephrology.

Table 5: Integrated summary of model performance, interpretability, and fairness. Both RF and XGBoost are accurate and clinically coherent, converge on renal biomarkers, and exhibit subgroup fairness.

Dimension	RF	XGBoost	Clinical Implications
Overall performance	High AUC (>0.95), well-calibrated	High AUC (>0.95), well-calibrated	Both reliable as predictive models
Key biomarkers	Urea, eGFR, Creatinine, Hemoglobin	Urea (dominant), Creatinine, Hemoglobin	Converges with nephrology knowledge
Local explanations	Balanced reliance on multiple features	Strong focus on few decisive features	RF for richer case narratives, XGBoost for fast triage
Fairness (Sex)	Stable AUC/sensitivity across male/female	Stable AUC/sensitivity across male/female	No sex-based bias detected
Fairness (Age)	Slightly lower specificity, calibration ≥50 yrs	Similar pattern: small drop ≥50 yrs	Reflects clinical heterogeneity, not bias
Interpretability style	Broader, multi-factor explanations	Sharper, concentrated explanations	Complementary perspectives for clinical use

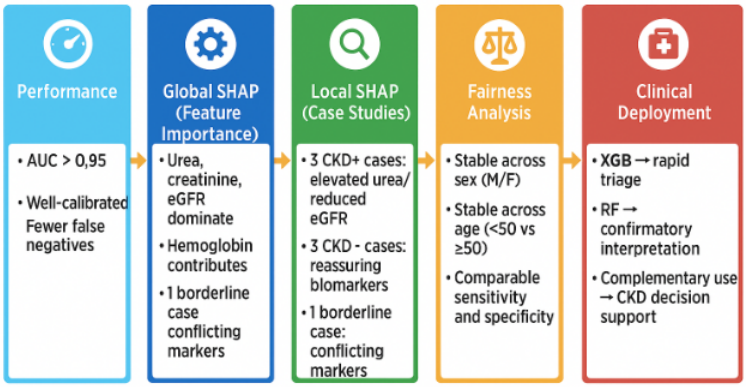


Figure 9: Integrated overview of findings. Both RF and XGBoost achieve high predictive performance, align with key renal biomarkers, provide case-level SHAP explanations, and maintain fairness across subgroups.

4.7 Economic Evaluation: Clinical Utility and Resource Impact

Although model accuracy and interpretability are essential for clinical adoption, screening programs must also demonstrate operational and economic value. To assess whether the RF and XGBoost models contribute meaningfully to CKD screening beyond predictive performance, we evaluated their clinical utility using decision-curve analysis (DCA) and examined their potential resource impact under a simulated screening scenario.

4.7.1 Decision-Curve Analysis (DCA). Decision Curve Analysis (DCA) helps answer a practical question: "If we use the model to decide who should get a confirmatory CKD test, does this lead to better outcomes than testing everyone or testing no one?"

Instead of looking only at accuracy, DCA evaluates whether a model leads to better real-world decisions, depending on the level of risk a clinician is willing to use as a cutoff. For example, some clinics may want to test patients only if their predicted risk is 20% or higher, while others may choose a higher or lower threshold depending on resources. See Figure 11.

In our results, both the Random Forest and XGBoost models improve decision-making across a wide range of thresholds. The Random Forest model performs especially well between 20% and 65% predicted risk—a range that aligns with typical screening practices. In this interval, the model helps: (i) identify more CKD cases early, and (ii) avoid unnecessary confirmatory tests for people who are unlikely to have CKD. This means the model is making testing decisions that are better than simple strategies such as testing everyone or testing no one.

XGBoost shows a similar pattern, although its performance changes more depending on the chosen risk threshold. At very low thresholds, the benefit of using a model becomes similar to simply testing everyone, which would not be practical in most healthcare settings because laboratory capacity is limited.

Overall, the DCA results show that both models can help clinicians choose who should receive follow-up CKD testing, with the Random Forest model offering the most stable and reliable improvement across common clinical decision ranges.

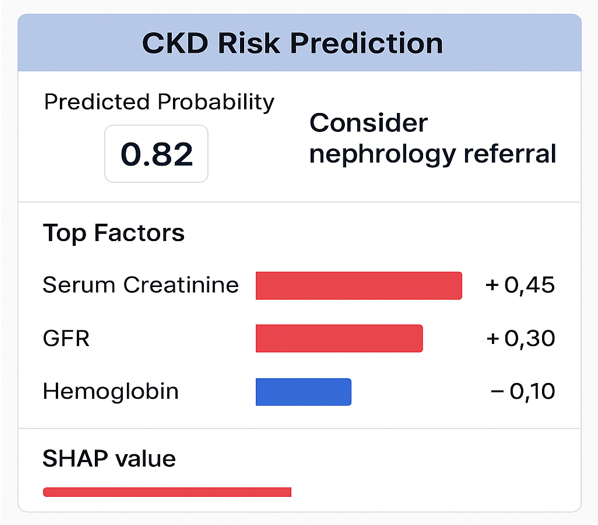


Figure 10: Screenshot of the prototype decision-support dashboard displaying a patient's risk score and associated SHAP feature contributions. For demonstration only; not clinically deployed.

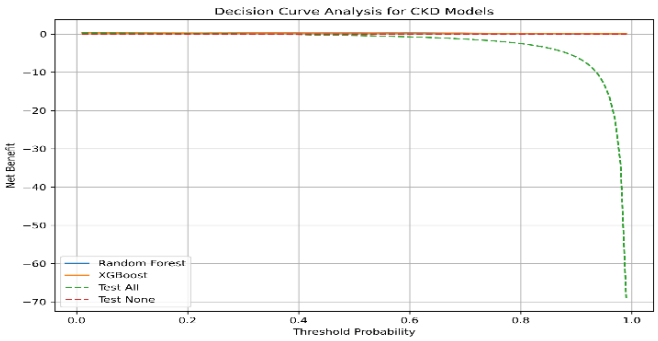


Figure 11: Decision Curve Analysis for CKD risk prediction models. Net benefit curves for the Random Forest and XGBoost models are shown across threshold probabilities from 0.01 to 0.99, alongside the reference strategies Test All and Test None.

4.7.2 Resource Impact Analysis. To make the potential economic impact of model-guided screening more concrete, we considered a simple example with 1,000 patients undergoing initial CKD risk assessment. In this hypothetical scenario, we assumed that:

- 10% of patients (100 out of 1,000) actually have CKD.
- A confirmatory CKD test costs USD 40.00 per person.
- Treating CKD detected late costs around USD 16,000.00 per patient.
- Patients with a predicted CKD risk of 40% or higher are sent for confirmatory testing.

For each model, we then counted how many patients would be:

- sent for confirmatory testing (positive model prediction),
- correctly identified early as having CKD (true positives), and
- missed and therefore detected late (false negatives).

Based on representative operating points at this threshold, we use the following illustrative values (see Table 5). The economic scenario presented below is illustrative and relies on simplified cost assumptions, as standardized CKD cost data for the studied healthcare context are not available. The purpose is to demonstrate the potential direction of economic impact rather than to provide definitive estimates

Two simple conclusions emerge from this example:

1. Early detection is the main driver of economic benefit. Most of the savings come from reducing the number of patients who reach late-stage CKD. Even modest differences in how many early cases a model identifies lead to meaningful differences in overall cost.
2. The cost of additional tests is very small compared with treatment costs.The money spent on confirmatory tests (thousands of dollars) is negligible relative to the treatment costs that are avoided (over one million dollars). This suggests that, in practice, accepting extra testing for some patients is economically reasonable if it leads to earlier diagnosis of CKD.

This simplified scenario does not replace a full health-economic evaluation, but it illustrates that model-assisted CKD screening has the potential to improve both patient outcomes and the efficient use of healthcare resources.

5 CONCLUSION

This study demonstrated that SHAP-based interpretability can strengthen CKD risk prediction by pairing high-performing machine learning models with transparent and clinically meaningful explanations. Random Forest and XGBoost both achieved strong discrimination and reliable calibration, while SHAP consistently attributed predictions to established renal biomarkers, aligning model behavior with nephrology expertise and supporting clinician confidence.

The work contributes three elements that are important for practical deployment: (1) integrated global and local SHAP analyses that provide complementary insights at the population and patient levels; (2) a fairness evaluation across age and sex to verify stable performance and explanations among key demographic groups; and (3) a prototype clinician-facing dashboard designed to embed interpretability directly into routine decision-making. The economic evaluation further illustrated that model-guided early identification may reduce unnecessary testing and support more efficient allocation of clinical resources.

Several limitations remain. The dataset originates from a single clinical setting, and interpretability has not yet been assessed through prospective studies involving real clinical users. Broader subgroup analyses and external validation across multi-center cohorts will be necessary to strengthen confidence in generalizability. Future work will incorporate iterative clinician feedback and compare SHAP with complementary interpretability approaches.

Overall, the findings indicate that combining calibrated prediction models with robust explanation techniques can provide a transparent, equitable, and clinically actionable foundation for CKD risk assessment. By uniting performance, interpretability, and fairness,

Table 6: Estimated resource impact for a 1,000-patient CKD screening scenario.

Model	Patients Sent for Testing	Early CKD Cases Found (True Positives)	CKD Cases Missed (False Negatives)	Costs Avoided by Early Detection (USD)*	Cost of Tests (USD)	Net Savings vs. Late Detection Only (USD)
RF	150	80	20	$80 \times 16{,}000 = 1{,}280{,}000$	$150 \times 40 = 6{,}000$	1,274,000
XGBoost	140	78	22	$78 \times 16{,}000 = 1{,}248{,}000$	$140 \times 40 = 5{,}600$	1,242,400

this work contributes to closing the gap between algorithmic prediction and practical medical decision support.

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